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Project Title:	Spray Cooling in Flooding Regime: Pool Boiling on Nano-Textured Surfaces		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
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Space Biology Element:	None		
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	The European Space Agency (ESA) recently initiated “Thermal Platform 1” research project, a ramification of DOLFIN, aiming research projects relevant for the work of the International Space Station (ISS) and for the further Space exploration. This document foresees that: "In the area of space applications and with the increasing size of space platforms or future exploration missions, large amounts of heat need to be dissipated by large space radiators. Efficient thermal control in space and the reduction of moving mechanical elements become of crucial importance and two-phase closed-loop systems can meet these requirements." The present project supported by NASA will address some of these issues and is planned to become an integral part of the ESA activities. The University of Illinois at Chicago (UIC) group supported by NASA in the framework of this project actively collaborated with the ESA supported group at the Technical University Darmstadt (Germany), where a group of 3 Master students, 1 PhD student, and 1 bachelor student		

Task Description:

were involved.
Drop and spray cooling in such situations could offer an attractive alternative, and our previous work, supported by NASA as a part of participation in DOLFIN, was directed on the enhancement of drop cooling using nano-textured surfaces. In spray cooling with surface flooding, heat transfer occurs in the regime of pool boiling. The regimes of pool boiling can be divided on the boiling curve as free convection boiling, nucleate boiling, transition boiling, and film boiling. Among them nucleate boiling is considered to be the most desirable heat transfer regime, since it is associated with high heat fluxes at surfaces with low superheat temperature, which is attractive for such practical applications as cooling of electronics, boilers, nuclear and chemical reactors, and refrigeration systems. However, there exists a critical value of heat flux called critical heat flux (CHF) at which nucleate boiling transforms into film boiling. The latter involves heat transfer through coolant vapor rather than through liquid coolant itself.

There were multiple efforts to increase CHF and boiling heat transfer coefficient (BHT), which is by definition the heat flux divided by the corresponding superheat temperature. In particular, boiling on roughened surfaces was found to increase BHT coefficient, which stems from an increased area density of nucleation sites. Microporous coatings to enhance pool boiling were also studied.

In our preliminary work on pool boiling on nano-textured surfaces under full gravity conditions, in the ground experiments the nano-textured surfaces used were copper platelets covered with copper-plated electrospun nanofibers. In addition, for comparison, pool boiling on the corresponding bare copper surfaces was experimentally studied. The results revealed that the heat flux and heat transfer coefficient in boiling on the nano-textured surfaces were about 3-8 times higher than those on the bare copper surfaces. This stems from the fact that nano-textured surfaces promote bubble growth by increasing the average temperature of fluid surrounding growing bubbles, as our experimental and theoretical results show. Hence, nano-textured surfaces manufactured from copper-plated nanofibers facilitate bubble growth rate and, thus, increase bubble detachment frequency. The critical superheat temperature of nano-textured surfaces was about 10 degrees C lower than that on smooth surfaces. In addition, the heat flux on the nano-textured surfaces in transition boiling was significantly higher than on the bare copper ones, since the presence of nanofibers prevented bubble merging and delayed formation of vapor film.

[ED. NOTE (12/2017): initial project from this work was " 'DOLFIN' DROP IMPACT ON POROUS NANO-TEXTURED SURFACES," (grant NNX10AR99G).]

[EDITOR'S NOTE February 2019: Work continues with grant NNX17AF33G, "Stationary and Swing-Like Pool Boiling for Microgravity Applications for Cooling of High- Power Microelectronics," Principal Investigator Alexander Yarin, PhD.]

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Heater surfaces were modified by depositing nanofibers. The resulting nano-textured surfaces facilitated a significant increase in heat removal and lower surface superheat in pool boiling. In particular, pool boiling of ethanol, water, and their binary mixtures on nano-textured surfaces comprised of copper-plated nanofibers was studied experimentally. The nanofiber-covered surfaces were formed using polymer nanofibers produced by the electrically-assisted supersonic solution blowing process followed by copper-plating. The pool boiling data on the nano-textured surfaces did not follow the standard boiling curve and showed a sharp deviation. Namely, the heat flux and accordingly, the heat transfer coefficient, were found to be significantly higher at low surface superheats. It was also demonstrated that the nano-textured surfaces developed in the present work are robust and do not deteriorate after several cycles of pool boiling experiments. These features are attractive for cooling of high-power microelectronics.
Also, to enhance heat transfer in forced convective boiling the mini-channel bottom was amended by nano-textured structures, which were periodic rectangular mats of electrospun polymer nanofibers. The fibers were about several hundreds of nanometer in diameter. The test fluid was FC-72. The flow in mini-channels contained trains of the Taylor bubbles. The role of the nanofibers was to retain the warm mini-channel bottom wetted, to prevent dry-out and thus to enhance the heat removal rate. It was shown experimentally that the time-average heat flux at the nanofiber-coated domains was 1.6 times higher than that at the uncoated ones. Accordingly, a significant decrease (by 5–8 K) in the superheat was observed. The heat transfer coefficient at the nanofiber mat-coated domains was found to be an order of magnitude higher than that at the uncoated domains. Such significant enhancement of heat transfer results from the fact that nanofiber mats facilitate wetting of the surface under the passing Taylor bubbles, thus delaying formation of vapor layer at the channel bottom. In order to get insights into the mechanisms of heat transfer enhancement invoked by the nanofiber coating, the evaporation of liquid in the vicinity of an apparent contact line of a single meniscus, where the liquid-vapor interface meets the nano-textured substrate, was investigated. The experiments have been performed for stationary, advancing, and receding menisci. It was found that the local heat flux in the vicinity of a stationary apparent contact line is increased by approximately 60% in the presence of nanofiber coating. The receding meniscus leaves behind an extended region, in which the pores within the nano-textured mat are fully or partially filled with evaporating liquid, which significantly contributes to the heat transfer enhancement. These phenomena were attributed to very strong capillary forces acting within the porous media and retaining the cooling liquid in contact with the heater surface.

In addition, pool boiling of Novec 7300 fluid and self-rewetting water-heptanol mixtures on bare copper surface and a copper surface coated with copper-plated nanofibers was studied experimentally. The experimental data revealed a significant increase in the heat removal rate up to the critical heat flux (CHF) on the copper-plated nanofiber surfaces in comparison with bare copper surfaces. Also, the critical heat flux increases on the copper-plated nanofiber surface, albeit it is reached at a lower surface superheat in comparison with bare copper surface. It was shown that prolong boiling in water facilitates oxidation of the layer of copper-plated nanofibers, and diminishes its roughness, albeit does not affect the heat transfer rate.

In a series of additional experiments pool boiling of Novec 7300 and DI water on bare copper surface and copper surface covered with supersonically-blown or electrospun polymer nanofibers was studied. In distinction from the previous stages of this project, nanofibers were not metal-plated. The elimination of the electroplating makes nano-texturing process extremely simple and applicable to a wide variety of surfaces. On the other hand, it comes with the price of a lower heat removal rate compared to that on metal-plated nanofibers, albeit the present experiments revealed that the reduction is not that drastic as it would be expected. It was also demonstrated that the supersonically-blown polymer nanofibers outperformed the electrospun nanofiber-coated surface. Supersonically-blown

nanofibers provide a larger number of nucleation sites than electrospun nanofibers or bare copper surface, and thus, facilitate nucleate boiling much stronger. The ultrafine supersonically-blown polymer nanofibers are very robust and did not delaminate from the heater surface either in Novec 7300 or DI water after prolonged vigorous boiling process. Overall, the enhancement of heat removal due to the ultrafine supersonically-blown polymer nanofibers on the heater surface in Novec 7300 was more significant than in DI water, which is associated with the smaller bubble size in the former case.

Then, problems with heat removal by means of pool boiling were addressed. It was demonstrated that heat removed in pool boiling from a heater mimicking high-power microelectronics could be facilitated by a swing-like motion of the heater. This swing-like motion could be beneficial for shedding a large vapor bubble which encapsulates high-power heaters in microgravity where buoyancy force is unavailable for vapor bubble removal. The swing-like motion is propelled by vapor bubble recoil, the force which exists irrespective of gravity and buoyancy. It was also demonstrated that this force could be significantly enhanced by depositing on the heater surface supersonically-blown polymer nanofibers with cross-sectional diameters below 100 nm. These nanofibers provide additional nucleation sites, resulting in much more frequent bubble nucleation and departure and thus a higher overall vapor recoil force propelling the heater motion. Such nanofibers strongly adhere to the heater surface and withstand prolonged harsh pool boiling. The measured velocity of the model swing-like heater in Novec 7300 fluid is about 1 cm/s.

[EDITOR'S NOTE February 2019: Work continues with grant NNX17AF33G, "Stationary and Swing-Like Pool Boiling for Microgravity Applications for Cooling of High- Power Microelectronics," Principal Investigator Alexander Yarin, PhD.]

Bibliography Type:	Description: (Last Updated: 06/13/2025)
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